

Band of Agents Hackathon 2026

Track 3: Regulated & High-Stakes Workflows

MV3 racing

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- **Project Title:** ComplianceFlow — Multi-Agent Financial Transaction Review Pipeline
- **Short Description:** 5 AI agents (Policy, Risk, Legal, Decision, Orchestra) coordinate through Band to automatically review financial transactions for AML/KYC compliance, producing audit-ready verdicts with full traceability.
- **GitHub:** <https://github.com/DNAK-dnak/Band-of-Agents-2026-Track-3>
- **Tags:** Band, Featherless API, LangGraph, compliance, finance, multi-agent

Objectives

- **High-Stakes Automation:** Designing agent systems that handle sensitive compliance and regulatory requirements without human oversight.
- **Rigorous Governance:** Ensuring every autonomous action follows explicit corporate policies, legal mandates, and safety guardrails.
- **Deterministic Execution:** Harnessing the coordination powers of the **Band Platform** to prevent deadlocks, race conditions, or hallucinated workflows.
- **Auditability & Transparency:** Maintaining a complete, immutable record of every decision, action, and communication for post-hoc review and accountability.

Problem Statement

Critical Vulnerabilities:

- **Regulatory Penalties:** Non-compliance with AML/KYC can result in billions in institutional fines (HSBC: \$1.9B, Deutsche Bank: \$629M).
- **Context Leakage:** Single LLM prompts struggle to balance overlapping regulatory frameworks simultaneously.

Technical Barriers:

- **State Management:** Keeping a reliable audit trail across distributed networks is notoriously complex.
- **Stuck Messages:** Network latencies often freeze async message pipelines, breaking the workflow.
- **Self-Mention Loops:** Agents can inadvertently trigger themselves, creating infinite processing cycles.

Proposed Solution

- **Separation of Concerns:** Isolated agents (*Policy, Risk, Legal, Decision*) each focus on a single domain, reducing prompt confusion and output variance.
- **Native Band REST API Integration:** Custom `agent_helper.py` module directly calls Band's message, event, and participant APIs — full control over the message lifecycle (processing → processed → failed) with automatic mention resolution and self-mention prevention.
- **Resilient LLM Inference:** `RetryingChatOpenAI` wrapper with exponential backoff handles Featherless AI rate limits gracefully. Agents automatically retry on 429/503 errors without human intervention.
- **Pipeline Supervision:** An Orchestra Agent monitors pipeline health in real time, detecting stalls, duplicates, and out-of-order handoffs before they become failures.

Workflow Design

1. Intake & Coordination

- The **Intake Coordinator** reads pending financial data sequentially from the `transactions.csv` file or user input.
- Creates a dedicated **Band Chatroom** per transaction, adds all agents via **Peer Discovery**, and submits the transaction to Policy Agent.

2. Sequential Multi-Agent Evaluation (The Pipeline)

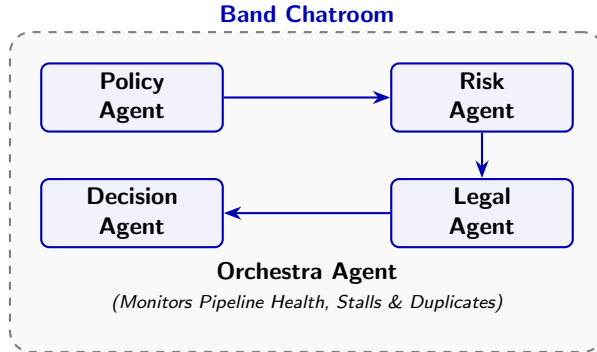
- Each specialized agent listens for @mentions, processes the transaction, and hands off to the next agent via Band's messaging API.
- Creates an automated chain-of-custody audit trail traversing from *Policy* → *Risk* → *Legal* → *Decision*.
- Orchestra Agent monitors the pipeline throughout, reporting health status.

3. Synthesis & Results Export

- The **Decision Agent** synthesizes all upstream assessments into a final verdict: AUTO-APPROVE, ENHANCED REVIEW, ESCALATE TO HUMAN, or DECLINE.
- The coordinator extracts the verdict and exports structured records to `results.csv`.
- The chatroom is torn down after completion to keep state clean.

System Pipeline & Chatroom Architecture

transactions.csv → **Coordinator** → **Band Chatroom** → **Pipeline** → results.csv



Agent Roles & Expected Outputs

Agent	Role	Output
Policy Agent	AML, KYC, sanctions screening, transaction monitoring	CLEAR / FLAGGED / BLOCKED
Risk Agent	Risk scoring, anomaly detection, counterparty analysis	LOW / MEDIUM / HIGH / CRITICAL
Legal Agent	BSA, OFAC, FATF compliance, filing requirements	APPROVED / CONDITIONAL / REQUIRES COUNSEL / BLOCKED
Decision Agent	Synthesizes all assessments, produces final recommendation	AUTO-APPROVE / DECLINE / ENHANCED REVIEW / ESCALATE
Orchestra Agent	Monitors pipeline health, detects stalls and duplicates	HEALTHY / STALLED / DUPLICATE

Band Platform Features Used

- **Peer Discovery:** Agents are found by name dynamically at runtime — eliminating hardcoded UUIDs.
- **Participant Management:** Automatically adds/removes agents based on individual room lifecycles.
- **Message Lifecycle:** Managed via a robust state machine (*processing* → *processed* → *failed*).
- **Thought Events:** Agents stream their internal reasoning before executing actions, providing a transparent audit trail.
- **REST API:** Programmatic control over room creation, teardown, message polling, and mention resolution.
- **Allowed Senders:** Deterministic routing where each agent only processes messages from its specific upstream neighbor, preventing loops.

- **Band:** Multi-agent coordination platform
- **Band REST API:** Native message lifecycle, participant management, peer discovery
- **Featherless AI:** Serverless LLM inference (Qwen 2.5 14B Instruct)
- **RetryingChatOpenAI:** Custom LLM wrapper with exponential backoff for rate limit resilience
- **LangChain:** Tool binding and structured tool execution
- **Python 3.11+:** Core development environment

- **55+ transactions** processed across real financial scandal scenarios
- **Test cases based on:** Enron, WorldCom, Wirecard, 1MDB, FTX, Lehman Brothers, Parmalat, Stanford Financial, Petrobras, Danske Bank
- **4 distinct verdict outcomes** produced based on transaction risk profiles
- **5 agents** connected to Band via SDK
- **Full audit trail** via thought events visible in Band chatroom
- **Automated end-to-end:** CSV intake → pipeline → results export
- **One-command launcher** for end-to-end workflow execution

What's next?

- **Dynamic Specialist Recruitment:** Decision Agent uses Band's peer discovery to recruit sanctions/PEP specialists on demand.
- **Sub-Room Delegation:** Legal Agent creates private rooms for sensitive findings, isolating classified data.
- **Parallel Assessment:** Risk and Legal agents run simultaneously instead of sequentially.
- **Google Sheets Integration:** Live transaction input from shared spreadsheets.
- **Agent Memory:** Using Band's Memory API for cross-session knowledge retention.